Provenance-Aware Belief Aggregation in Multi-Agent Communication Under Convention Heterogeneity

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Abstract

The standard move in multi-agent systems is to weight messages by source reliability. We argue this answers the wrong question whenever reliability and communication convention are distinct latent variables. When agents report through *source-specific conventions*, mappings from internal readings to messages that can invert the signal or transform it depending on context, the real difficulty is interpreting the message, and no reliability weight can fix a misinterpretation. We prove two structural limitations for the convention class studied here. First, no aggregation rule restricted to nonnegative source weights can implement the Bayes-optimal decision when an inverted source's Bayes weight exceeds the realized combined contribution of the remaining sources on some message vector (the hypothesis of Theorem 1); the inverted source's own reliability strengthens the condition, the other sources' works against it. Second, no rule assigning a single context-independent scalar weight per source can reproduce a source's Bayes log-odds contribution when its convention flips across observable contexts (any stability $\rho > 0$); when $\rho > 1/2$ the effective sign itself flips with context, and such rules fail on decisions as well (Theorem 2). We complement these with a consistency result: a Bayesian aggregator that infers each source's latent convention from a support set concentrates on the observational-equivalence class of the truth as evidence accumulates, so its predictive log-odds converge to the oracle's. We introduce a finite-grid Bayesian latent-convention aggregator and evaluate it on a benchmark of eight involutive conventions. Bayesian aggregation consistently outperforms the scalar baselines and matches or exceeds the learned calibrators we tested (each in a single, untuned configuration) given identical support information—a transformer meta-learner with 4.1× the attention baseline's parameters finishes within half a point, a small but statistically detectable deficit, while a decoder-privileged diagnostic nearly reaches the oracle—with the clearest margins at high convention stability and in larger groups. In multi-agent consensus it improves group accuracy by 37.5 percentage points over naive averaging (twelve agents), finishing within 4.3 points of an oracle. We also report calibration (negative log-likelihood, Brier score, expected calibration error): the Bayesian posterior scores best probabilistically but grows overconfident as conventions stabilise, a limitation. A new stability-boundary experiment maps when per-source decoding helps versus hurts: at low convention stability ($\rho = 0.5$) a population-average mapping beats per-source Bayesian decoding (paired +0.0196 at $K = 10$), with the advantage shrinking as support grows and vanishing at $K = 50$; at $\rho \geq 0.6$ decoding wins decisively. A post-repair validation programme adds three further evidence types: controlled panels reproducing the mechanism with a dose–response in measured heterogeneity; real human judgments (CrowdTruth medical relation tasks) in which per-source behaviour measurably flips across contexts (37% of workers sign-flip between relation families) and a calibrated context-decoding estimator beats the best calibrated scalar on accuracy, log loss and calibration error across two tasks; and an adversarial suite pricing the assumptions—under a convention switch between support and query, every support-based method collapses to chance. On CIFAR-10H, majority vote is already near ceiling and source modelling gains nothing—a boundary condition. We make no claim about multi-agent communication in general.

Keywords

multi-agent systems, source calibration, trust aggregation, latent-variable inference, collective decision-making

1 Introduction

Consider a deceptively simple sentence: *"The bridge is safe."* Heard from a structural engineer, it licenses crossing. Heard from a five-year-old, it licenses nothing. Heard from an anonymous account, it licenses suspicion. The message is identical; the decision it supports is not. Something about the *source* changes what the message means.

Multi-agent systems meet this problem at scale. Agents exchange beliefs, recommendations, sensor readings, and every receiver must decide how much weight each message deserves. A substantial class of trust and reputation models represents source differences primarily through reliability or trust parameters, formalised as $\hat{y} = \text{sign}(\sum_s w_s X_s)$ with $w_s \geq 0$. Weighted voting, reliability-weighted averaging, trust-weighted fusion. The nonnegative weight is everywhere. We argue that reliability weighting is insufficient for this setting.

The failure is in how messages are interpreted. Take a source that is perfectly reliable, its internal reading always matching the truth, but which reports through a private convention: it flips its reading, or XORs it with an observable context bit. Its messages are uninterpretable without the convention, and no nonnegative weight, however carefully estimated, repairs the aggregation. Reliability and convention are orthogonal. A source can be maximally trustworthy and still unreadable. We call the resulting problem *convention heterogeneity*. It is distinct from the reliability heterogeneity the literature has studied.

By *provenance-aware* aggregation we mean rules that condition on source identity. The decision-theoretic value of provenance is the change in decision value from knowing who sent the message: with the source known, the receiver can decode through that

source's convention; without it, the message may be uninterpretable. Our impossibility theorems characterise regimes where source identity carries this decision value yet no scalar reliability rule can extract it, and our aggregator is a constructive method for extracting it.

We make five contributions.

(1) We formalise the problem as a generative latent-variable model in which source identity is part of the process that generates each message, with involutive conventions, a stability parameter, and reliability kept strictly separate.
(2) On the impossibility side, nonnegative source weights cannot implement the Bayes-optimal rule under inverted conventions whenever the inverted source's weight dominates the rest (the hypothesis of Theorem 1).
(3) We prove that no context-independent scalar weight can reproduce a source's Bayes log-odds contribution under context-dependent conventions at any stability $\rho > 0$, and that for $\rho > 1/2$ — where the source's effective sign flips with context — such rules also fail to implement the Bayes decision (Theorem 2).
(4) A Bayesian aggregator that infers each source's latent convention concentrates on the observational-equivalence class of the truth as evidence accumulates (Theorem 3).
(5) Finally, the empirical piece: a finite-grid Bayesian latent-convention aggregator that infers each source's convention, stability, and reliability from a labelled support set, evaluated on a benchmark of eight involutive conventions in nine lettered experiments (A–I), an LLM-communication benchmark, and a real-data study.

The Bayesian aggregator consistently outperforms the scalar baselines under the synthetic benchmark, with the clearest margins at high convention stability and in larger groups, and matches or exceeds the learned calibrators we tested given identical support information: a transformer meta-learner finishes within half a point of it, and only a diagnostic with privileged decoded features nearly reaches the oracle. (Bayes leads the learned MetaAttn at every group size tested, by 1.4–3.5 points; see §6.7.)

Two scope notes. The benchmark is synthetic, the convention family finite, the statistics exploratory: Experiment A uses 100 seeds; Experiments B, C, E, and I use 30; Experiments D, F, G, and H use 10. The implemented posterior grid covers stability $\rho \in \{0.5, \ldots, 1.0\}$ only, so Theorem 3's on-grid hypothesis — and every empirical number — should be read with that restriction. Our claim is conditional but sharp: reliability and communication conventions are distinct latent variables, and aggregation methods that model only reliability cannot recover Bayes-optimal decisions when the convention changes what a message means.

2 Related Work

2.1 Trust and reputation in multi-agent systems

The dominant paradigm for handling source differences is trust and reputation, and for this lineage the scalar characterization is fair. TRAVOS [33] estimates the probability that a source provides accurate information from its track record; the Beta reputation system [18] maintains Bayesian posteriors over trustworthiness; FIRE [17], REGRET [29], and BLADE [28] extend the idea to witness reputation, social structure, and learned accuracy models. Each of these models, in the end, *how often* a source is right. That is the gap we work in: a source can be maximally reliable under every one of these frameworks and still be unreadable, because reliability and convention are different things.

2.2 Opinion dynamics and social learning

Classical opinion dynamics, DeGroot [11] and Friedkin–Johnsen [14], aggregate through nonnegative influence weights: exactly the class our Theorem 1 rules out for inverted conventions. Bayesian social learning [1] and naive learning in networks [15] ask when networks converge to the truth, but take the mapping from state to signal as fixed and common knowledge. We study what happens when that mapping is private, heterogeneous, and has to be inferred.

2.3 Crowdsourcing, truth discovery, and label noise

The closest technical relatives of our model live here, and honesty requires placing the model inside that lineage rather than beside it. The founding model is Dawid and Skene's [10]: each observer is characterised by a full confusion matrix, estimated by EM — there is no scalar weight anywhere in it. The lineage that followed is a sequence of structured per-annotator response models: GLAD [35] (per-annotator expertise interacting with item difficulty), Raykar et al. [27] (per-annotator sensitivity and specificity), Raykar and Yu [26] (who note that annotators with sensitivity plus specificity below one are negatively informative and can be reversed by EM — prior art, in the context-free case, for our inverted sources), IBCC [19] (Bayesian classifier combination with Dirichlet priors over the confusion matrices), and, inside truth discovery proper, Zhao et al. [39] (a Bayesian generative model with distinct per-source false-positive and false-negative rates), alongside the truth-discovery methods of [12, 23, 38]. The label-noise literature makes the same object central: a per-class transition matrix, estimated or inverted to correct the loss [22, 24].

Per (source, context), our (c, ρ, r) model induces exactly a binary symmetric channel — a strict special case of one such transition matrix, as Section 3 derives. We claim no new model class, and the scalar-weight characterization of our introduction applies to the trust lineage above and to weight-based opinion dynamics, not to this response-model literature. What that literature does not do is threefold. First, none of its response models is conditioned on an *observable context* that changes the mapping itself: the confusion matrix is a property of the annotator, not of the (annotator, context) pair. Second, none ties the per-context channels through a finite family of mappings shared across sources — the tying that makes estimation from $K = 10$ labelled examples per source possible here, and a restriction a reader should price in. Third, none proves an impossibility statement — that context-independent scalar aggregation cannot represent the Bayes rule once the mapping varies with observable context (Theorem 2) — or characterizes the stability boundary at which per-source decoding stops paying (Experiment I). Our contribution is that framing, the cross-context tying as

a small-K inductive bias, and the boundary — not a new response model. Truth discovery asks who to trust. We are asking what they meant, in contexts where what they meant changes with the context.

2.4 Robust and adversarial aggregation

Byzantine-tolerant aggregation [5] and Byzantine-robust distributed learning [37] handle sources that actively try to corrupt the outcome. Our setting is different: nobody is attacking anyone. The sources are honest and often highly reliable; they simply speak in different conventions, so there is no attacker to detect. What remains is a decoding problem, and reliability scores cannot see it.

2.5 Signaling, conventions, and pragmatics

The idea that a fixed signal can carry opposite conventional meanings is classical. Lewis's analysis of convention [20] and Skyrms's evolutionary account [30] study signaling systems in which the same signal is associated with different states under different equilibria — our direct and inverted mappings are the two-element case — and cheap-talk theory [8] studies senders whose messages must be decoded through known biases. The framing of our model also grew out of partial information decomposition [4, 36]: does source identity carry information beyond message content? What survived is decision-theoretic — our theorems target decision rules directly, since statistically real information ($I(X;S \mid M) > 0$) need not imply decision value. The intuition is old in pragmatics: Gricean implicature [16] and relevance theory [32] treat speaker identity and context as constitutive of meaning; our conventions are a minimal formal version of that insight.

2.6 Multi-agent LLM systems

Recent work on LLM agents has explored debate protocols [13, 21], round-table deliberation [7], multi-agent evaluation [6], and layered aggregation of proposals [34]. These frameworks model source differences through confidence scores, voting, or an aggregator model's judgement, and empirical studies find debate gains sensitive to protocol design and not always robust over simpler baselines [31]. What they generally do not model is the possibility that agents encode readings through source-specific, context-dependent conventions. Our contribution relative to this literature is the impossibility results: Theorems 1 and 2 show that when the mapping from readings to messages itself varies by source and context, the thing to learn is a decoder, not a weight.

3 Problem Formulation

The key modelling choice is that *source identity is part of the generative process of the message*, not a label attached to it afterwards: each source encodes its internal reading through a source-specific *convention* before transmitting, so what the receiver observes, and what it means, depends on who sent it and in what context.

3.1 Generative model

Let $Y \in \{0,1\}$ be the binary truth with uniform prior, and let there be N sources. Each source s is described by three parameters, and keeping them separate is the entire point of the model:

- A **convention** $c_s \in C$: an involutive function $f_{c_s} : \{0,1\} \times \{0,1\}^d \to \{0,1\}$ mapping the source's internal reading and an observable context $a \in \{0,1\}^d$ to a message. Involution means $f_c(f_c(v,a),a) = v$: applying the convention twice recovers the reading, which is what makes decoding possible at all.
- A **stability** $\rho_s \in [0,1]$: the probability that the source applies its convention on any given transmission rather than reporting its reading directly.
- A **reliability** $r_s \in (1/2, 1)$: the probability that the source's internal reading equals the truth. This is the only quantity the trust literature estimates.

On each round, source s draws reading $R_s = Y$ with probability r_s (else $1 - Y$), transmits $M_s = f_{c_s}(R_s, a)$ with probability ρ_s (else R_s), and the receiver observes the message M_s together with the context a.

We work with the **centered message** $X_s = 2M_s - 1 \in \{-1, +1\}$, so that aggregation is sign-symmetric. Define the log-odds weight $\lambda_s = \log(r_s/(1-r_s)) > 0$ and, for inversion-only conventions, the convention sign $\sigma_s = +1$ for honest (identity) conventions and $\sigma_s = -1$ for inverted conventions.

The full generative process for one round, with source parameters $\theta = \{(c_s, \rho_s, r_s)\}_{s=1}^N$ fixed, is:

$$Y \sim \text{Bernoulli}(1/2) \tag{1}$$

$$A_s \sim P_A \quad \text{independently per source and round} \tag{2}$$

$$R_s \mid Y: \quad P(R_s = Y \mid Y) = r_s \tag{3}$$

$$B_s \sim \text{Bernoulli}(\rho_s) \tag{4}$$

$$M_s = \begin{cases} f_{c_s}(R_s, A_s), & B_s = 1 \\ R_s, & B_s = 0 \end{cases} \tag{5}$$

The receiver observes (M_s, A_s) for each source. A **support set** of K labelled triples $\{(M_s^{(k)}, A_s^{(k)}, Y^{(k)})\}_{k=1}^K$ per source is available for calibration; test queries are fresh draws from the same process with the same source parameters.

Relation to transition-matrix models. Fix a source s and a context a, and write the convention in XOR form $f_{c_s}(v,a) = v \oplus g_{c_s}(a)$ (every convention in our family has this form; only 8 of the 16 Boolean functions g of two bits occur). The induced channel $Y \mapsto M_s$ is then a binary symmetric channel with error rate

$$\varepsilon_a = \begin{cases} 1 - r_s, & g_{c_s}(a) = 0, \\ (1-\rho_s)(1-r_s) + \rho_s r_s, & g_{c_s}(a) = 1, \end{cases}$$

since in an unflipped context both branches transmit the reading, while in a flipped context the direct branch errs with probability $1 - r_s$ and the convention branch with probability r_s. Per (source, context) the model is therefore a strict special case of the per-source, per-context transition matrices estimated in the crowdsourcing literature (Section 2): an unrestricted model has a free error rate for each (source, context) pair, while ours ties all contexts through the two scalars (ρ_s, r_s) and one of eight flip patterns. The signed effective weight in context a is $\lambda_a = \log \frac{1-\varepsilon_a}{\varepsilon_a}$: it equals $+\lambda_s$ in unflipped contexts, 0 at $\rho_s = 1/2$ in flipped contexts, and $-\lambda_s$ only at $\rho_s = 1$. Theorem 2 is a statement about this quantity.

3.2 Context assumptions

The context A_s is a d-bit vector ($d = 2$ in all experiments), drawn i.i.d. from a fixed distribution P_A independently per source, per round, and independently of the truth Y. It is observed without noise, and the support and test phases share the same P_A. No source chooses or influences the context. These assumptions matter for Theorem 2: P_A must put positive mass on the convention's distinguishing inputs.

Edge cases. With $K = 0$ the posterior of Section 5 equals the prior and the predictive is the prior mixture; nothing in Sections 4–5 requires $K \geq 1$ except the learned calibrators of Section 6, which is why all experiments use $K \geq 1$. Theorem 3's predictive claim holds for P_A-almost every query context. The parameter ranges are open ($r_s > 1/2$, $\rho_s > 0$) by design: at $r_s = 1/2$ a source carries no information, and at $\rho_s = 0$ its convention is never applied, so its contexts are indistinguishable.

3.3 Aggregation classes

We compare two classes of aggregation rules. The **nonnegative-weight class**

$$\mathcal{A}_+ = \left\{ \hat{y}_w(X) = \operatorname{sign}\left(\sum_s w_s X_s \right) : w_s \geq 0 \right\} \tag{6}$$

captures reliability weighting: each source contributes its (centered) message scaled by a nonnegative trust weight. The **signed class**

$$\mathcal{A} = \left\{ \hat{y}_v(X) = \operatorname{sign}\left(\sum_s v_s X_s \right) : v_s \in \mathbb{R} \right\} \tag{7}$$

additionally allows negative weights, i.e. flipping a source's contribution. Both classes assign a single scalar per source, independent of context. In both classes, $\operatorname{sign}(0)$ is resolved by an independent fair coin, which makes the $\frac{1}{2}$ tie credit in Lemma 1 exact.

Oracle. The oracle knows the ground-truth parameters (c_s, ρ_s, r_s) for every source and applies the Bayes-optimal aggregation rule directly. No learning, no inference. It is a reference upper bound, not a competitor.

4 Structural Limitations

4.1 Nonnegative weights cannot invert a source

Theorem 1 (Nonnegative-weight insufficiency). *Suppose sources use inversion-only conventions with full stability ($\rho_s = 1$ for all s), and at least one source j uses an inverted convention ($\sigma_j = -1$). Suppose there exists a message vector X_{-j} from the remaining sources such that $\left|\sum_{s \neq j} \sigma_s \lambda_s X_s\right| < \lambda_j$. Then no rule in $\mathcal{A}_+$ agrees with the Bayes rule on all message vectors. Here $\mathcal{A}_+$ is the class of rules $\hat{y}(X) = \operatorname{sign}(\sum_s w_s X_s)$ with $w_s \geq 0$ fixed per source at decision time: the weights may be estimated from support data by any procedure, but they cannot depend on the current message vector or context.*

Proof. Consider the inversion-only regime, where the Bayes log-odds are $\mathrm{LO}(X) = \sum_s \sigma_s \lambda_s X_s$. Fix X_{-j} satisfying the hypothesis and form the pair $X^{(1)} = (X_{-j}, -1)$, $X^{(2)} = (X_{-j}, +1)$, differing only in source j's message. Writing $S = \sum_{s \neq j} \sigma_s \lambda_s X_s$,

$$\mathrm{LO}(X^{(1)}) = S + \lambda_j > 0 \quad \text{and} \quad \mathrm{LO}(X^{(2)}) = S - \lambda_j < 0,$$

using $\sigma_j = -1$ and $|S| < \lambda_j$. Hence the Bayes decision is strictly *decreasing* in X_j on this pair: $+1$ then -1. Both log-odds are nonzero (the inequalities are strict), so the fair-coin tie convention of Section 3 never engages on this pair.

Any rule $U \in \mathcal{A}_+$ computes $U(X) = \operatorname{sign}(\sum_s w_s X_s)$ with $w_s \geq 0$. Moving from $X^{(1)}$ to $X^{(2)}$ changes the score by $2w_j \geq 0$, so U is *non-decreasing* in X_j and cannot reproduce the Bayes flip from $+1$ to -1. Contradiction. □

The argument is a monotonicity obstruction, and it is worth stating plainly what it rules out: the Bayes rule sometimes has to move *against* a source's message, and no nonnegative weighting can ever do that. The hypothesis is sufficient rather than generic. It names concrete configurations where the failure bites, which is what an impossibility result is for. The cleanest witness is the single-source case: with X_{-j} empty, $S = 0 < \lambda_j$ holds trivially, so a lone inverted source already defeats every nonnegative rule.

Corollary 1. *In the inversion-only regime with full stability ($\rho_s = 1$), the signed class $\mathcal{A}$ contains the Bayes rule, attained at $v_s = \sigma_s \lambda_s$.*

4.2 Signed scalars cannot handle context-dependence

Allowing negative weights fixes inversion. It does not fix the deeper problem: when a source's convention conditions on context, the source's Bayes log-odds contribution is itself a function of the context — shrinking, vanishing, or flipping sign as stability varies — and no single scalar can represent a function.

Theorem 2 (Scalar insufficiency for context-dependent conventions). *Let source j have convention c with flip function g_c (so $f_c(v, a) = v \oplus g_c(a)$), stability $\rho_j \in (0, 1]$ and reliability $r_j \in (1/2, 1)$. Suppose there exist contexts $a^{(1)} \neq a^{(2)}$ in the support of P_A with $g_c(a^{(1)}) = 0$ and $g_c(a^{(2)}) = 1$. Write $\lambda_j = \log \frac{r_j}{1 - r_j}$ and let ε_a be the induced channel error of Section 3.*

*(a) [*Log-odds*] Source j's Bayes log-odds contribution for message m in context a is $\ell_j(m, a) = (2m - 1) \log \frac{1 - \varepsilon_a}{\varepsilon_a}$. In context $a^{(1)}$ the weight is $+\lambda_j$; in context $a^{(2)}$ it is a strictly decreasing function of ρ_j, equal to $+\lambda_j$ at $\rho_j = 0$, to 0 at $\rho_j = 1/2$, and to $-\lambda_j$ at $\rho_j = 1$. In particular, whenever $\rho_j > 0$ the contribution differs across the two contexts, so no context-independent scalar w_j — of either sign — equals source j's Bayes log-odds contribution in both.*

*(b) [*Decisions*] If moreover $\rho_j > 1/2$, so that the $a^{(2)}$-weight $\ell_j^{(2)}$ is negative while the $a^{(1)}$-weight is positive, fix the other sources' messages and contexts so that their total Bayes log-odds S satisfies $|S| < \min(\lambda_j, |\ell_j^{(2)}|)$ (attainable, e.g. source j alone, where $S = 0$). Then there exist observations $x^{(1)} = (X_{-j}, X_j, a^{(1)})$ and $x^{(2)} = (X_{-j}, X_j, a^{(2)})$ sharing the same centered message vector $X = (X_{-j}, X_j)$, with $X_j = 2M_j - 1$, on which the Bayes decisions differ. Consequently no context-independent function of the message vector alone — in particular, no scalar-weight rule in $\mathcal{A}$ or $\mathcal{A}_+$ — implements the Bayes rule on all observations. A context-dependent weight $w_j(a)$ is a different function class, and is not ruled out.*

Proof. See Appendix A. □

Remark 1. *Two boundary cases delimit the result. At $\rho_j = 1/2$ a flipped context contributes exactly zero log-odds: the source is not*

misread there, it is silent, and no scalar is contradicted. At $\rho_j < 1/2$ the flipped-context weight is positive but shrunk, so part (a) still rules out exact scalar representation of the log-odds, while decisions need not flip; part (b)'s sign flip needs $\rho_j > 1/2$, and the full-stability picture — the weight flipping from $+\lambda_j$ to $-\lambda_j$ — is the endpoint $\rho_j = 1$. Two assumptions deserve to be explicit. First, the receiver observes *the context a; without that, context-dependent conventions are trivially undecodable. Second, the convention family is* known *to the receiver; the theorem is about representational capacity, not discovery. The obstacle is context-dependence, not sign.*

4.3 Equal-reliability optimum

Lemma 1 quantifies the price of the monotonicity obstruction: with p honest and q inverted sources at reliability r and full stability, the Bayes accuracy counts all $p + q$ decoded readings, while the best unsigned rule attains only the honest-only analogue, i.e. it optimally discards the inverted sources entirely (statement, proof, and a numerical illustration are in Appendix A).

4.4 The Bayesian aggregator is consistent

The results above say what *fixed* rules cannot do. The natural question is whether the Bayesian aggregator of Section 5, which learns the convention instead of fixing a weight, actually finds the truth. It does, in the standard sense.

Theorem 3 (Posterior consistency). *Fix a source with true parameters $\theta^* = (c^*, \rho^*, r^*)$ lying on the finite grid $C \times$ RhoGrid $\times$ RGrid. Let $[\theta^*]$ be its observational-equivalence class: the grid points inducing the same distribution over support triples (M, a, Y). Then, as $K \to \infty$,*

$$P([\theta^*] \mid \mathcal{D}_s) \xrightarrow{a.s.} 1,$$

and the aggregator's predictive log-odds converge almost surely to the oracle log-odds (exact Bayes inference with known θ^*).

Proof. See Appendix A. □

Remark 2. *The theorem promises class concentration, not point identification, and the distinction is load-bearing: on our own grid, distinct parameters can be observationally identical. Under an inversion-only convention, (ρ, r) enter the triple distribution only through the agreement probability $p = \rho(1-r) + (1-\rho)r$, so $(\rho, r) = (1.0, 0.55)$ and $(0.6, 0.75)$—both on-grid—induce identical support-triple distributions; under the identity convention, ρ drops out of the likelihood entirely. What the posterior identifies is the triple distribution itself, and that is all the predictive needs: the query likelihood is the same conditional, constant across each equivalence class, so the predictive log-odds converge to the oracle's regardless.*

A caveat on scope: the theorem assumes θ^* lies exactly on the grid, while our simulations draw ρ and r continuously. Off-grid, Berk's theorem [3] gives concentration on the KL-closest grid point, but that point is unique only up to predictive equivalence: where two grid points induce the same support-triple distribution — as in the inversion example above — the posterior may split mass between them rather than settle on one, and this splitting occurs in our own runs. The predictive claim is unaffected, because every such point induces the same query likelihood, so the predictive log-odds converge to those of the KL-closest predictive distribution. The grid is fine enough ($\Delta\rho = 0.1, \Delta r = 0.05$) that the projection is close to the truth. We cite Experiment C (Section 6) — at $K = 50$ the aggregator reaches 0.8204 ± 0.0029 (30 seeds), essentially the expected reliability ceiling $\mathbb{E}[r] = 0.825$ — as evidence about the grid's accuracy only, not as part of the consistency claim.

Algorithm 1 Grid-posterior aggregation for one source

Require: support set $\mathcal{D}_s = \{(M_i, a_i, Y_i)\}_{i=1}^K$; query (M_q, a_q)

for each cell (c, ρ, r) in $C \times \{0.5, \ldots, 1.0\} \times \{0.55, \ldots, 0.95\}$ **do**

  $\ell(c, \rho, r) \leftarrow \sum_{i=1}^K \log\left[(1-r) + (2r-1)q_i(c, \rho)\right]$

end for

$P(c, \rho, r \mid \mathcal{D}_s) \leftarrow \mathrm{softmax}(\ell)$ ▷ uniform prior

$\mathrm{LO} \leftarrow \log \dfrac{\sum_{c,\rho,r} P(c, \rho, r \mid \mathcal{D}_s)\, P(M_q \mid Y{=}1, c, \rho, r)}{\sum_{c,\rho,r} P(c, \rho, r \mid \mathcal{D}_s)\, P(M_q \mid Y{=}0, c, \rho, r)}$

return $\hat{Y} = \mathbb{1}[\mathrm{LO} > 0]$ ▷ ties resolve to 0 in the implementation

Corollary 2 (Joint aggregator). *Under the conditional independence of Section 3, the joint aggregator that sums per-source predictive log-odds inherits Theorem 3: if every source's parameters lie on the grid, the summed predictive log-odds converge almost surely to the joint oracle log-odds as each $K_s \to \infty$.*

Proof. Each source's predictive log-odds converge a.s. to its oracle log-odds by Theorem 3; a finite sum of almost-surely convergent terms converges almost surely, and the sum of the per-source oracle log-odds is the joint oracle log-odds under conditional independence. □

5 Bayesian Latent-Convention Aggregator

The theorems say aggregation has to decode messages, not merely weight them. So we infer each source's latent parameters from a labelled *support set* $\mathcal{D}_s = \{(M_i, a_i, Y_i)\}_{i=1}^K$ of past (message, context, truth) triples, and aggregate the resulting predictive log-odds.

Priors. Independent uniform priors over the finite convention family C and over the discretised stability and reliability grids:

$$c \sim \mathrm{Uniform}(C), \quad \rho \sim \mathrm{Uniform}(\{0.5, 0.6, \ldots, 1.0\}),$$

$$r \sim \mathrm{Uniform}(\{0.55, 0.60, \ldots, 0.95\}).$$

The prior is discrete over exactly the grid the posterior is enumerated on, so there is no continuous-to-discrete mismatch.

Likelihood. For a support example, let $A_i = \mathbb{1}[f_c(M_i, a_i) = Y_i]$ indicate that decoding the message through convention c recovers the truth, and $B_i = \mathbb{1}[M_i = Y_i]$ indicate direct agreement. With $q_i = \rho A_i + (1-\rho)B_i$ — a convex combination of the two observed indicators, not itself a probability: the per-example probability that the transmitted symbol matches the truth is $(1-r) + (2r-1)\, q_i$, as the likelihood shows —

$$P(\mathcal{D}_s \mid c, \rho, r) = \prod_{i=1}^K \left[(1-r) + (2r-1)\, q_i\right]. \tag{8}$$

Posterior. The posterior $P(c, \rho, r \mid \mathcal{D}_s)$ is computed by exact enumeration over the finite grid $\rho \in \{0.5, 0.6, \ldots, 1.0\}$ and $r \in \{0.55, 0.60, \ldots, 0.95\}$. This is a numerical grid approximation, not a closed-form posterior, and we make no claim to the contrary.

Prediction. For a query (M_q, a_q), the predictive log-odds are the log of the ratio of posterior-averaged likelihoods:

$$\mathrm{LO} = \log \frac{\sum_{c,\rho,r} P(c,\rho,r \mid \mathcal{D}_s)\, P(M_q \mid Y=1,c,\rho,r)}{\sum_{c,\rho,r} P(c,\rho,r \mid \mathcal{D}_s)\, P(M_q \mid Y=0,c,\rho,r)}, \tag{9}$$

and the aggregator predicts $\hat{Y} = 1$ iff $\mathrm{LO} > 0$. Note this is *not* the posterior average of the parameter-conditional log-odds. The two differ by Jensen's inequality, and only the form above is the exact Bayesian predictive. In the multi-agent setting, per-source predictive log-odds are summed, which is exact under the conditional-independence assumption of Section 3; the oracle below is defined there as well. One implementation detail: the code resolves exact ties ($\mathrm{LO} = 0$) deterministically to $\hat{Y} = 0$, whereas the fair-coin convention of Section 3 is used only in the theory of Section 4 (Lemma 1), where the $\frac{1}{2}$ tie credit is part of the statement. Exact ties are isolated configurations except at $\rho = 0.5$ in flipped contexts, where a single source's log-odds contribution vanishes identically; all reported numbers use the implementation's rule.

6 Experiments

6.1 Benchmark design

The benchmark uses eight involutive conventions over two context bits (a_1, a_2): identity, NOT, and six XOR-with-context variants (full table in Appendix B). Each method receives a support set of K labelled (message, context, truth) triples per source and is evaluated on held-out queries. Experiment A uses 100 seeds; Experiments B, C, E, and I use 30; Experiments D, F, G, and H use 10 seeds (0–9); and the LLM benchmark uses 20 trials. Unless a caption says otherwise we report mean ± 95% CI (t-interval over seeds or trials); the Experiment D and F tables report ±1 std, as their captions state. The statistics are exploratory: enough to separate stable gaps from noise, not confirmatory tests.

Baselines. *NoSupport* (content-only); *UnsignedScalar* (nonnegative weight, testing Theorem 1); *SignedScalar* (signed weight, testing Theorem 2); *ContextConditioned* (Laplace context-matching); *CtxLogReg* (per-source context-dependent logistic weights); *MetaMean* and *MetaAttn* (DeepSets-style neural calibrators); *DeepAttn* (wider attention decoder, $d = 256$); *TransformerMeta* (self-attention meta-learner over the support set); *ContextMatchFreq* (nonparametric context-matching fallback; named `BetaBinomial` in the code and in earlier drafts); *LogOddsScalar* (per-source log-odds scalar; §6.4); *Bayes* (the finite-grid aggregator of Section 5, with *BayesUniv*/*BayesFam* variants in the K-sweep); *NaiveMean* and *MajorityVote* (uncalibrated group rules); and the *Oracle* (true (c_s, ρ_s, r_s), Bayes-optimal rule).

6.2 Convention stability sweep

We first vary the convention stability ρ from 0.5 (conventions rarely applied) to 1.0 (always applied), with $K = 10$ and $r \in [0.7, 0.95]$ (Table 1).

At $\rho = 0.5$ the picture is mixed, and precision about what vanishes matters. NoSupport — which ignores the *support set*, while still consuming each query's message and contexts — beats Bayes (0.6485 ± 0.0029 vs. 0.6317 ± 0.0023; paired $\Delta = +0.0169 \pm 0.0028$, $d_z = +1.19$). The messages are not empty at $\rho = 0.5$: in contexts where a source's convention does not flip, a message still agrees with the truth with probability r; only in flipped contexts does the agreement probability collapse to $1/2$ (Section 3). What vanishes at this support size is per-source *decodability*: the population-average mapping of §6.3 exploits the residual signal without paying the per-source estimation cost, and wins on bias–variance grounds; §6.3 maps the crossover directly. As $\rho \to 1$ the picture changes: NoSupport collapses toward chance (0.4831), unsigned and signed scalars stall, while the Bayesian aggregator climbs monotonically to 0.7576 ± 0.0021 and the neural calibrators partially recover (0.7012–0.7170). The price of ignoring conventions grows with their stability. The nonparametric ContextMatchFreq trails Bayes at every ρ (at $\rho = 1.0$: 0.6763 ± 0.0020, paired $\Delta = -0.0812 \pm 0.0020$, $d_z = -8.10$): flexibility without the generative structure does not recover the lost accuracy. (A comparability note: NoSupport here and ContentOnly in the §6.4 ablation are the same rule family measured in different runs — this sweep trains on streams with $\rho \in (0.5, 1.0)$, the ablation on $\rho \in (0.6, 1.0)$ — so their absolute levels are not directly comparable; the analytic population mapping of §6.3, which involves no training, matches NoSupport at $\rho = 0.5$ (0.6496 vs. 0.6485).) The per-query expected reliability ceiling under full convention stability is $\mathbb{E}[r] = 0.825$ even for the oracle; the consensus experiment (§6.7) shows how independent sources combine to lift accuracy far above that ceiling.

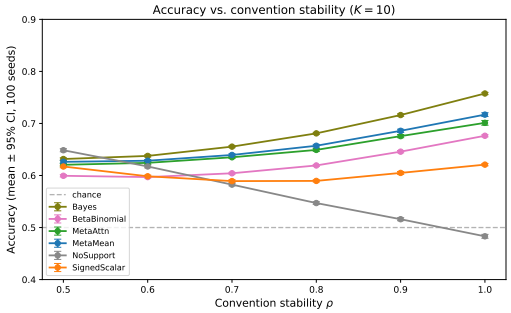


Figure 1: Accuracy vs. convention stability ρ ($K = 10$). Error bars show 95% CI over 100 seeds.

6.3 The stability boundary: when decoding hurts

The $\rho = 0.5$ boundary condition in the stability sweep raises a prescriptive question: when conventions are only weakly stable, is per-source decoding ever worth its estimation cost? We characterise the boundary directly. Define the *population mapping* as the Bayes-optimal content-only rule: for each (m, a_1, a_2) predict $\arg\max_y P(Y = y \mid m, a_1, a_2)$ where the probability averages over the training population (c uniform over eight conventions, $\rho \sim \mathrm{Uniform}(0.5, 1)$, $r \sim \mathrm{Uniform}(0.7, 0.95)$). This is the analytic equivalent of what NoSupport learns; it matches the banked NoSupport accuracy at $\rho = 0.5$ (0.6496 vs. 0.6485) without any training. The comparison is then clean: per-source Bayesian decoding (which estimates a convention per source from K support triples) versus applying the population-average mapping (which estimates nothing per source).

Figure 2 shows the crossover at $K = 10$ is sharp. At $\rho = 0.5$ the population mapping beats per-source Bayes by +0.0196 (paired

Table 1: Accuracy vs. convention stability ρ ($K = 10$); mean ± 95% CI over 100 seeds.

ρ	Bayes	ContextMatchFreq	MetaAttn	MetaMean	NoSupport	Signed	Unsigned
0.5	0.6317 ± 0.0023	0.5994 ± 0.0022	0.6206 ± 0.0028	0.6262 ± 0.0026	0.6485 ± 0.0029	0.6173 ± 0.0023	0.6273 ± 0.0020
0.6	0.6376 ± 0.0021	0.5969 ± 0.0019	0.6243 ± 0.0027	0.6284 ± 0.0022	0.6174 ± 0.0026	0.5985 ± 0.0023	0.6032 ± 0.0021
0.7	0.6554 ± 0.0019	0.6043 ± 0.0018	0.6350 ± 0.0022	0.6394 ± 0.0023	0.5825 ± 0.0023	0.5892 ± 0.0025	0.5823 ± 0.0022
0.8	0.6809 ± 0.0019	0.6193 ± 0.0018	0.6492 ± 0.0027	0.6571 ± 0.0026	0.5471 ± 0.0025	0.5896 ± 0.0021	0.5675 ± 0.0022
0.9	0.7161 ± 0.0023	0.6458 ± 0.0022	0.6756 ± 0.0030	0.6859 ± 0.0031	0.5160 ± 0.0029	0.6049 ± 0.0027	0.5610 ± 0.0023
1.0	0.7576 ± 0.0021	0.6763 ± 0.0020	0.7012 ± 0.0046	0.7170 ± 0.0037	0.4831 ± 0.0037	0.6209 ± 0.0022	0.5526 ± 0.0023

95% CI $[+0.0151, +0.0241]$, $d_z = +1.63$): estimating a per-source convention from $K = 10$ examples adds estimation noise that outweighs the decodable signal at this stability. At $\rho = 0.6$ the ordering already flips (-0.0174, CI $[-0.0214, -0.0134]$, $d_z = -1.64$), and the decoding margin grows rapidly with stability (-0.2803 at $\rho = 1.0$, $d_z = -12.74$). The heatmap (Figure 3) shows the same pattern across support sizes: the population mapping wins only in the low-stability regime, and its advantage shrinks as K grows. At $\rho = 0.5$ the paired gain falls from +0.0286 at $K = 2$ to +0.0196 at $K = 10$, +0.0065 at $K = 20$, and vanishes at $K = 50$ (+0.0001, CI $[-0.0034, +0.0037]$): with enough per-source data, decoding catches up even when conventions are unstable.

The practical rule is therefore conditional on both stability and data: decode per-source when conventions are at all stable ($\rho \gtrsim 0.6$) or support is ample; fall back to the population mapping when conventions are unstable *and* support is scarce. The boundary is not a failure of decoding but a bias–variance tradeoff: per-source convention inference is high-variance, and when the signal is weak the variance dominates.

A natural refinement is to *select* per source: decode when a stability test on the source's own support set passes, else fall back to the population mapping. We ran exactly that selector (30 seeds, $K = 10$; it chooses per-source Bayes for a fraction of sources rising from 0.60 at $\rho = 0.5$ to 0.95 at $\rho = 1.0$). It does not beat the better fixed rule in either regime: at $\rho = 0.5$ it scores 0.640 in its own runs, between per-source Bayes (0.631) and the population mapping (0.652); at $\rho = 1.0$ it scores 0.749 against Bayes's 0.758. Selection on the same scarce support data does not come free, so the boundary above is stated for the two fixed rules.

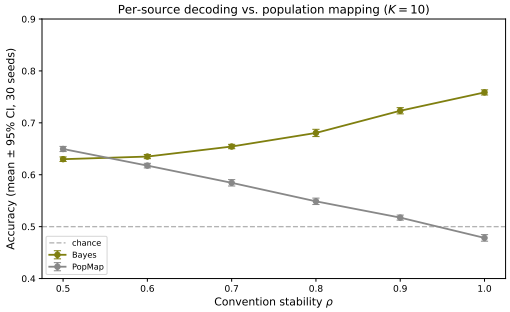


Figure 2: Per-source Bayesian decoding vs. the population mapping ($K = 10$). Error bars show 95% CI over 30 seeds. The crossover is sharp: the population mapping wins only at $\rho = 0.5$.

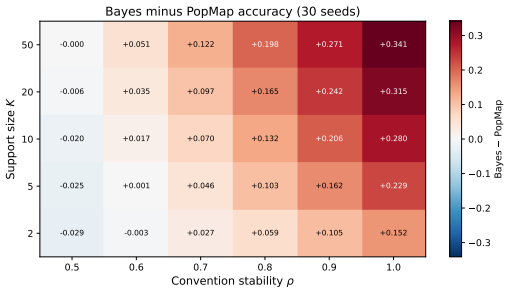


Figure 3: Bayes minus population-mapping accuracy over the $\rho \times K$ grid (30 seeds). Blue: decoding helps; red: the population mapping helps. The advantage of the population mapping is confined to low stability and shrinks with K.

6.4 Representation ablation

At $\rho \approx 1.0$ and $K = 10$, accuracy follows the hierarchy predicted by the theorems (Table 2, Figure 4): content-only near chance, unsigned scalar low, signed scalar at 0.6207, context-conditioned scalar at 0.7350, the attention calibrator at 0.7238, the Bayesian aggregator at 0.7590, approaching the oracle at 0.8263.

Table 2: Representation ablation at $\rho \approx 1$, $K = 10$; mean ± 95% CI over 30 seeds. Methods above the rule use only the support set available to every method; Dec and DecCtx below the rule receive ground-truth decoded features at inference and are diagnostics, not competitors.

Method	Accuracy
Deployable methods (support set only)	
ContentOnly	0.5396 ± 0.0116
UnsignedScalar	0.5497 ± 0.0027
SignedScalar	0.6207 ± 0.0041
ContextConditioned	0.7350 ± 0.0048
AttentionCalibrator	0.7238 ± 0.0053
ContextMatchFreq	0.6780 ± 0.0036
DeepAttn	0.7299 ± 0.0040
TransformerMeta	0.7544 ± 0.0047
Bayes	0.7590 ± 0.0044
Oracle	0.8263 ± 0.0028
Privileged diagnostics (ground-truth decoded features)	
MetaAttn+Dec	0.8240 ± 0.0032
MetaAttn+DecCtx	0.8240 ± 0.0032

The table reads as a hierarchy that mirrors the theory: raw messages near chance, unsigned scalars low (Theorem 1), signed

scalars at 0.6207, context-conditioned scalars at 0.7350 (Theorem 2), full latent-convention inference at 0.7590, and the oracle at 0.8263. Two notes. First, our initial ContextConditioned baseline silently collapsed to SignedScalar; we reimplemented it with Laplace-smoothed context-matching support agreement. Second, the diagnostic MetaAttn+Dec variants receive the true-convention-decoded bit at inference and reach the oracle ceiling (0.8240 vs. 0.8263), so the learned calibrator's remaining gap is representational, not a capacity problem. Finally, the hand-crafted ContextConditioned (0.7350 ± 0.0048) and the higher-capacity AttentionCalibrator (0.7238 ± 0.0053) have overlapping 95% CIs, so the observed point-estimate ordering is not clearly separated at this sample size.

A disclosure about the scalar baselines' functional form: UnsignedScalar and SignedScalar as implemented form a one-parameter sub-family — each source enters through its support agreement rate, scaled by a single learned global multiplier (linear in the agreement rate, not its log-odds), signed in the second case. The strongest signed member of the richer per-source log-odds family — weight $\log(\hat{p}/(1-\hat{p}))$ per source, parameter-free — is decision-identical to SignedScalar on all 30 seeds (both 0.6207 ± 0.0041), so the scalar-class conclusion does not rest on the restricted form.

We also tested stronger learned baselines, since a sufficiently expressive source × context meta-learner might match Bayes without generative machinery. DeepAttn, a wider attention decoder ($d = 256$), reaches 0.7299 ± 0.0040—no better than the smaller MetaAttn, so capacity is not the bottleneck. TransformerMeta, a self-attention meta-learner over the support set (104,834 parameters, against MetaAttn's 25,667 at the same width $d = 64$ — not capacity-matched), reaches 0.7544 ± 0.0047. The paired gap to Bayes is small but statistically detectable ($\Delta = -0.0046$, 95% CI $[-0.0070, -0.0022]$, $d_z = -0.71$): we report it as a half-point deficit, not a tie, applying the same paired-CI standard here as to the smaller gaps in this paper. A sufficiently expressive sequence model can nearly match the Bayesian decoder here, but none of the learned methods we tried beats it. ContextMatchFreq, a nonparametric context-matching fallback, trails badly at 0.6780 ± 0.0036 (paired $\Delta = -0.0810 \pm 0.0036$, $d_z = -8.30$): flexibility without the generative structure does not recover the lost accuracy. The decoder structure does the work, not capacity alone.

CtxLogReg tests the same objection from the discriminative side (10-seed run): per-source logistic regression on the same $K = 10$ support set, with the logit a context-dependent weight on the message, $w_0 + m\,(v_0 + v_1 a_1 + v_2 a_2 + v_{12} a_1 a_2)$, which nests ContextConditioned's per-context sign and can express any of the eight conventions' decoding maps. It reaches only 0.637 ± 0.007, below ContextConditioned and Bayes with non-overlapping CIs. The result is not a regularisation artefact ($\lambda = 0.01$ and 0.1 agree at $K = 10$). Accuracy does rise with support in a dedicated sweep ($\lambda = 0.1$, 10 seeds): 0.534 at $K = 2$, 0.590 at $K = 5$, 0.637 at $K = 10$, 0.677 at $K = 20$, 0.713 at $K = 50$ — more data helps the discriminative fit, without closing the gap to the structured methods at matched K. At $K = 10$, expressiveness without the right inductive bias does not help.

A further ablation isolates *why* Bayes beats the scalar baselines, and answers the paper's central empirical question: does posterior uncertainty matter beyond point-estimate decoding? We decompose the gap on the Experiment B streams (30 seeds; all comparisons paired; an earlier 10-seed sensitivity run gives the same ordering). The posterior's own MAP plug-in — decide with the oracle log-odds formula at the grid posterior's MAP cell — reaches 0.7530 ± 0.0048: full posterior averaging is worth only 0.6 points over the posterior's own point estimate (paired $\Delta = +0.0060$, 95% CI $[+0.0031, +0.0090]$). A posterior-mean plug-in (posterior means of ρ, r under the MAP convention) does slightly worse, 0.7494 ± 0.0052. EMDecoder, which fits the same generative model by EM and decides with the MAP convention, reaches only 0.6870 ± 0.0054 — and the decomposition shows the deficit is mostly not about point estimates at all. EMDecoder's decision rule drops the fitted stability $\hat{\rho}$ from the plug-in weight; restoring it (the exact oracle formula at EM's estimates) recovers 0.7343 ± 0.0038, a paired gain of 4.7 points (95% CI $[+0.0438, +0.0508]$) attributable to the rule. The remainder is fit quality: EM's MAP convention is correct for 0.393 ± 0.014 of sources at $K = 10$ (5,400 sources), against 0.640 ± 0.014 for the grid posterior's MAP on the same streams. Bayes's 7.2-point margin over EMDecoder therefore decomposes as ≈ 4.7 points of decision rule, ≈ 1.9 points of convention recovery, and ≈ 0.6 points of posterior averaging. Explicit context conditioning recovers much of the lost performance; Bayesian inference over the latent convention adds the rest, mostly by recovering conventions better than EM's point fit does.

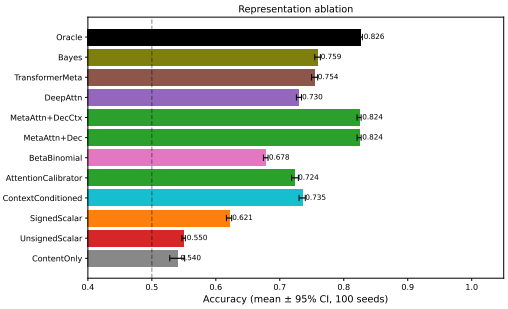


Figure 4: Representation ablation at $\rho \approx 1$, $K = 10$. Error bars show 95% CI over 30 seeds. Unsigned scalars fail (Theorem 1); signed scalars fix inversion but not context-dependence (Theorem 2).

6.5 Support-size scaling on held-out conventions

Test conventions $\{4, 5, 6, 7\}$; $K \in \{1, 2, 5, 10, 20, 50\}$ (support sets are nonempty, as the neural encoders require). "Held-out" means the following. The neural calibrators are trained on sources using conventions $\{0, 1, 2, 3\}$ and evaluated on sources using conventions $\{4, 5, 6, 7\}$, with K labelled support examples per source available at test time for calibration. BayesUniv considers all eight conventions as candidates. BayesFam restricts its candidate set to $\{4, 5, 6, 7\}$: it knows the test family, and the comparison between the two Bayesian variants therefore measures the value of that knowledge rather than generalisation to unseen conventions.

Table 3: Accuracy on held-out conventions vs. support size K; mean ± 95% CI over 30 seeds.

K	BayesUniv	BayesFam	ContextMatchFreq	MetaAttn	MetaMean	NoSupport
1	0.5865 ± 0.0052	0.6423 ± 0.0053	0.4981 ± 0.0025	0.5448 ± 0.0058	0.5335 ± 0.0069	0.4789 ± 0.0121
2	0.6446 ± 0.0052	0.6880 ± 0.0051	0.5333 ± 0.0030	0.5627 ± 0.0057	0.5588 ± 0.0045	0.4854 ± 0.0131
5	0.7118 ± 0.0049	0.7422 ± 0.0049	0.5968 ± 0.0035	0.5931 ± 0.0061	0.5822 ± 0.0072	0.4813 ± 0.0144
10	0.7626 ± 0.0036	0.7824 ± 0.0039	0.6660 ± 0.0038	0.6131 ± 0.0075	0.5975 ± 0.0071	0.4860 ± 0.0126
20	0.7911 ± 0.0032	0.8043 ± 0.0030	0.7260 ± 0.0036	0.6302 ± 0.0089	0.6116 ± 0.0079	0.4870 ± 0.0126
50	0.8176 ± 0.0030	0.8204 ± 0.0029	0.7921 ± 0.0035	0.6427 ± 0.0106	0.6257 ± 0.0104	0.4832 ± 0.0127

Both Bayesian variants rise monotonically with K, converging to approximately 0.82 at $K = 50$ (0.8204 ± 0.0029), which is, up to noise, the information ceiling $\mathbb{E}[r] = 0.825$. The neural calibrators climb from chance to 0.63–0.64 and then flatline, a gap of roughly 18 percentage points at $K = 50$ under the tested conditions. ContextMatchFreq is the surprise second-best here: 0.7921 ± 0.0035 at $K = 50$, well above the neural methods, though still clearly below both Bayesian variants. The wider DeepAttn calibrator, present in the run files but omitted from the table, plateaus alongside MetaAttn (0.641 at $K = 50$), so the plateau is not specific to the smaller attention model. We do not claim this bounds learned calibrators in general; with matched support information and modest capacity, structure beats learning here.

[Figure omitted for space; see Appendix B.]

6.6 Reliability heterogeneity

$\rho \in [0.99, 1.0]$. The three regimes are *mean-matched*: homogeneous $r \in [0.79, 0.81]$, moderate $r \in [0.70, 0.90]$, heterogeneous $r \in [0.65, 0.95]$, all with $\mathbb{E}[r] = 0.80$. This matching matters: with the means matched, any remaining difference across regimes is attributable to the *spread* of reliabilities alone.

Table 4: Bayesian robustness to reliability heterogeneity (mean-matched, $\mathbb{E}[r] = 0.80$); mean ± std over 10 seeds.

Regime	Bayes	MetaAttn	NoSupport
Homogeneous	**0.728 ± 0.013**	0.699 ± 0.012	0.596 ± 0.009
Moderate	**0.727 ± 0.017**	0.707 ± 0.011	0.589 ± 0.010
Heterogeneous	**0.731 ± 0.011**	0.706 ± 0.009	0.593 ± 0.006

Bayes outperforms the baselines evaluated in this experiment across all regimes, and its accuracy is flat at ≈ 0.73 across spreads (all three 95% confidence intervals overlap). An ablation pinning reliability to its mean ($r = 0.80$, convention and stability still inferred) reproduces this flat profile exactly ($\Delta \leq 0.002$ in every regime), so the flatness should not be attributed to per-source reliability inference: it reflects robustness of convention identification to reliability misspecification. (The per-query ceiling is $\mathbb{E}[r] = 0.80$; the remaining gap is the usual cost of inferring parameters from $K = 10$.)

[Figure omitted for space; see Appendix B.]

6.7 Multi-agent consensus

N agents observe a shared binary truth, each with its own (c, ρ, r) sampled i.i.d. as follows: $c_s \sim \text{Uniform}(C)$, $\rho_s \sim \text{Uniform}([0.7, 1.0])$, $r_s \sim \text{Uniform}([0.7, 0.95])$. The sampling distribution does not depend on N; realized diversity increases with N because more i.i.d. draws are combined.

Table 5: Group accuracy vs. number of agents under mixed conventions; mean ± 95% CI over 30 seeds.

N	Naive	MetaAttn	Bayes	Oracle
2	0.5189 ± 0.0208	0.7350 ± 0.0194	0.7700 ± 0.0159	0.8228 ± 0.0174
3	0.5428 ± 0.0286	0.7750 ± 0.0210	0.7967 ± 0.0207	0.8839 ± 0.0193
4	0.5439 ± 0.0204	0.8283 ± 0.0194	0.8428 ± 0.0188	0.9128 ± 0.0150
6	0.5494 ± 0.0247	0.8583 ± 0.0180	0.8761 ± 0.0127	0.9450 ± 0.0087
8	0.5811 ± 0.0256	0.8894 ± 0.0169	0.9183 ± 0.0135	0.9678 ± 0.0101
12	0.5739 ± 0.0197	0.9239 ± 0.0173	**0.9489 ± 0.0114**	0.9917 ± 0.0039

Naive averaging sits near chance at every N: its sign *is* the majority vote, so the two coincide and we report a single column. Both calibrated methods rise sharply with N; at $N = 12$, Bayes is within 4.3 points of the oracle (0.9489 vs. 0.9917) and 37.5 points above naive averaging (0.5739). Bayes leads the learned MetaAttn at every group size, by 1.4 to 3.5 points; the paired 95% CIs exclude zero throughout, but with Cohen's d_z between −0.39 and −0.73 these are modest effects, not decisive separations. As N grows the naive-to-calibrated gap widens: more agents help only if you can decode them.

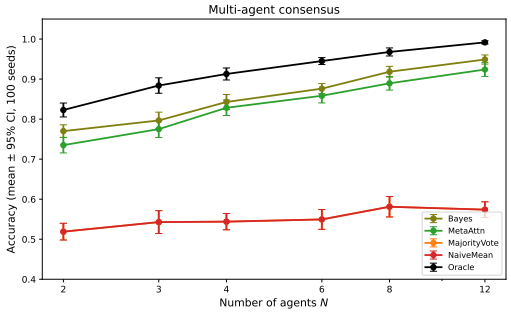


Figure 5: Multi-agent consensus accuracy vs. number of agents. Error bars show 95% CI over 30 seeds.

6.8 Calibration

Accuracy is only half the story: the Bayesian aggregator returns probabilities, so the re-run experiments also record negative log-likelihood, Brier score, and expected calibration error (ECE, 10 equal-width bins). In the stability sweep (Figure 6), Bayes has the best NLL and Brier score at every ρ (at $\rho = 1.0$: NLL 0.5480 ± 0.0018, Brier 0.1820 ± 0.0008). Its ECE, however, rises with stability, from 0.0170 ± 0.0012 at $\rho = 0.5$ to 0.0836 ± 0.0020 at $\rho = 1.0$: as conventions stabilise, the posterior concentrates and grows overconfident. At $\rho = 1.0$ the learned MetaAttn is actually the better-calibrated of the two (ECE 0.0587 ± 0.0024)—the Bayesian posterior's overconfidence at high stability is a genuine weakness, not shared by the discriminative calibrator.

In the ablation (Experiment B) the oracle is the best calibrated by far (ECE 0.0105 ± 0.0012), with Bayes at 0.0837 ± 0.0044. In consensus, the two methods' calibration moves in opposite directions as the group grows: Bayes's ECE falls from 0.1359±0.0108 at $N = 2$ to 0.0614±0.0044 at $N = 12$—more independent sources genuinely sharpen the posterior—while MetaAttn's ECE rises steeply from

0.1443 ± 0.0112 to 0.3389 ± 0.0160. The learned aggregator's probabilities degrade as the group grows even while its accuracy improves. Calibration is thus a second axis on which the methods differ, and a real limitation of the current Bayesian aggregator: without recalibration, its probabilities should not be taken at face value at high convention stability.

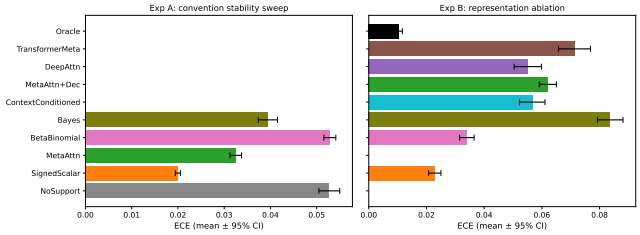


Figure 6: Expected calibration error (mean ± 95% CI) in the stability sweep (left) and the ablation (right). Bayes is the best-scoring probabilistically but grows overconfident as conventions stabilise; in consensus (not shown) its ECE falls with group size while the learned calibrator's rises.

6.9 Candidate-set misspecification

In practice the modeller never knows the full convention family, so we stress-test a wrong candidate set: sources drawn from all eight conventions while the Bayesian candidate set is restricted to $\{0, 1, 2, 3\}$ (*BayesHalf*); *BayesFull* (all eight) is the well-specified control. $\rho \in [0.99, 1.0]$, $r \in [0.7, 0.95]$, $K = 10$. We report overall accuracy plus the split by whether the true convention was inside the candidate set.

Table 6: Accuracy under candidate-set misspecification; mean ± std over 10 seeds. *BayesHalf* splits accuracy by whether the true convention was inside its candidate set.

Method	Accuracy
BayesFull	0.756 ± 0.013
BayesHalf	0.685 ± 0.012
in-candidate ($c \in \{0..3\}$)	0.767 ± 0.015
out-of-candidate ($c \in \{4..7\}$)	0.599 ± 0.020
MetaAttn	0.731 ± 0.010
Oracle	0.824 ± 0.007

Performance decreases from 0.756 to 0.685 overall, with the reduction concentrated on sources whose true convention lies outside the candidate set. On in-candidate sources the restricted model matches the well-specified control (0.767 vs. 0.756, within noise); on out-of-candidate sources accuracy falls to 0.599, still well above chance, since the posterior spreads mass over the nearest available conventions rather than committing to a single wrong one. Overall BayesHalf drops 7 points (0.685 vs. 0.756). Notably, the learned MetaAttn (0.731) lands between the two Bayesian variants: with no convention family assumed, it handles misspecification better than the restricted Bayesian but worse than the well-specified one. The Bayesian aggregator is a closed-world method, strong inside its assumed family; this experiment bounds the cost of getting the family wrong.

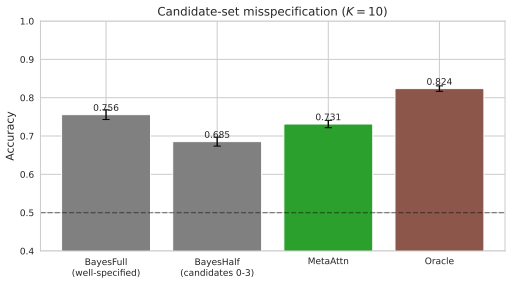


Figure 7: Accuracy under candidate-set misspecification ($K = 10$). Error bars show ±1 std over 10 seeds.

6.10 Correlated sources: breaking the independence assumption

Every result so far keeps the model correctly specified. Here it is deliberately *not* the data-generating process: the Bayesian aggregator assumes sources are conditionally independent given the truth, so we correlate their errors. Ten sources (five pairs) report on one shared truth per query; within-pair query-time errors correlate at ≈ 0.5 (Gaussian copula, calibrated; measured 0.44–0.52), with ExpB draws otherwise ($\rho \in [0.99, 1.0]$, $r \in [0.7, 0.95]$, $K = 10$). Support data stay independent, isolating the misspecification to the combination step. A paired-seed independence arm is the control; MetaAttn could not be re-run here (torch unavailable). Correlation costs every method about a point, *including the oracle*, which also sums independent log-odds: Bayes 0.971 → 0.958, ContextConditioned 0.962 → 0.950, SignedScalar 0.847 → 0.844, naive 0.449 → 0.457, Oracle 0.994 → 0.983 (mean ± 95% CI over 10 seeds; paired per-seed deltas −0.013, −0.012, −0.004, +0.008, −0.011, all within noise of each other). The problem itself gets harder under correlation, and Bayes is not disproportionately hurt: the ranking Bayes > ContextConditioned > SignedScalar > naive is preserved in both arms.

6.11 LLM communication benchmark

As a step beyond simulation, we instantiate the five-source population with a real language model. Each source is an independent `gpt-4o-mini` session given one of five induced conventions: A answers truthfully; B negates every belief; C negates exactly when a random context flag is 1; D answers truthfully but carelessly (about 15% of answers flipped); E answers truthfully on science/nature questions and negates everything else. The model is *told* its private belief on each proposition (correct with probability 0.85), so the experiment tests convention-following and decoding, not the model's factual knowledge: a controlled communication benchmark, not a claim about real-world deployment. We use 120 verifiable propositions, $K = 8$ labelled support answers per source and $Q = 20$ queries per trial, over 20 trials; the Bayesian aggregator reuses the synthetic convention family directly on the parsed Yes/No bits. Its candidate set in the released code is $\{0, 1, 2, 3, 8\}$, which cannot express source E's topic-conditioned rule: E's mapping is a function of the proposition's topic, not of the context bits the synthetic family flips on.

Decoding conventions helps enormously. We report this run as a *pre-fix-harness* result: the numbers below were banked before the harness defects described in the caveats were found; the banked "oracle" was handed an inverted rule for source E and is beaten by every decoder, so it is not an upper bound and we do not treat it as one; and the numbers are not reproducible from the released (fixed) code. Raw API outputs from the original run were not retained; the released harness now saves them. Mean accuracy ± 95% t-CI over the 20 trials: SignedScalar 0.9950 ± 0.0072, ContextConditioned 0.9900 ± 0.0096, Bayes 0.9875 ± 0.0104, nominal Oracle 0.9800 ± 0.0140, and NaiveMean 0.8175 ± 0.0451. Paired gains over the naive mean are +0.1775 ± 0.0477 (d_z = 1.74), +0.1725 ± 0.0483 (d_z = 1.67), +0.1700 ± 0.0477 (d_z = 1.67) and +0.1625 ± 0.0491 (d_z = 1.55) respectively: about seventeen points, with every interval well clear of zero. Near-ceiling accuracies leave little headroom between the decoders themselves, so this experiment separates decoding from not decoding rather than ranking the decoders.

Two caveats. First, convention adherence (fraction of support messages matching the induced rule) was high for A–D (0.925, 0.950, 0.938, 0.888), but the diagnostic for E read 0.100. Auditing the code showed the benchmark's expected rule for E was inverted relative to its prompt: it checked negate-on-science where the prompt says truthful-on-science. Under the inverted check a perfectly compliant source would read near zero — not exactly zero, since the model's own errors blur the reading — so 0.100 is consistent with high compliance with the prompt as written; it does not pin compliance at any precise figure such as 90%. The failure was in our harness, not the model, and the mapping is fixed in the released code. Because E's true rule was also missing from every decoder's candidate set, the decoders solved a harder problem than the design intended; the direction of the resulting bias on the decoder gaps is not determined, and we do not claim the gaps are conservative. Second, the private beliefs here are supplied, so this says nothing about extracting beliefs the model actually holds; it validates the aggregation machinery given convention-following speakers.

6.12 Real human judgments

As a check on genuine human data, we aggregate the CIFAR-10H judgments [25]: roughly 51 category labels for each of the 10,000 CIFAR-10 test images from 2,571 annotators (514,200 judgments after attention checks are removed; every annotator labels exactly 200 images, and all 2,571 workers had estimation-half observations). We split the images into disjoint estimation and evaluation halves of 5,000, fit per-worker confusion matrices on the estimation half with the CIFAR-10 gold labels, and aggregate each evaluation image's annotations to predict its gold label. Five rules are compared: MajorityVote, accuracy-weighted voting (scalar reliability weights), a Bayesian rule using plug-in smoothed confusion matrices (BayesDirichlet; no integration over matrix uncertainty), Dawid–Skene EM [10] warm-started from the estimation-half confusion counts and iterated to convergence — the label posterior is recomputed from the current confusion matrices at every iteration (5 iterations in the full regime, 92 in the five-judgment regime) — and an oracle using confusion matrices fit on all data. Mean accuracy ± 95% CI over the 5,000 evaluation images, with paired deltas against majority vote reported below:

Method	~51 judgments	5 judgments
MajorityVote	0.9924 ± 0.0024	0.9844 ± 0.0034
AccuracyWeighted	0.9926 ± 0.0024	0.9846 ± 0.0034
BayesDirichlet	0.9930 ± 0.0023	0.9842 ± 0.0035
DawidSkene	0.9928 ± 0.0023	0.9850 ± 0.0034
Oracle	0.9960 ± 0.0017	0.9878 ± 0.0030

Paired deltas against majority vote (mean ± 95% CI) are tiny: with all judgments, +0.0002±0.0004 (accuracy-weighted), +0.0006± 0.0010 (BayesDirichlet), +0.0004±0.0008 (DawidSkene), and +0.0036± 0.0017 (oracle). With five judgments the deltas are +0.0002±0.0012, −0.0002 ± 0.0014, +0.0006 ± 0.0014, and +0.0034 ± 0.0017.

This is nearly a null result, and that is informative. CIFAR-10H annotators are individually strong (mean accuracy 0.949), so majority vote already reaches 0.9924, and no source-modelling method adds anything measurable: every paired interval covers zero. Even the oracle, with perfect knowledge of every worker, gains only about a third of a point. The five-judgment regime—the realistic crowdsourcing budget—tells the same story at 0.9844. This is the boundary condition the thesis predicts rather than a failure of it: the large synthetic gains (Experiment E's +37.5 points over naive averaging) come from heterogeneous, unreliable sources, and these annotators are neither. The experiment's value is in showing the machinery does not manufacture an advantage where none exists.

Two scope notes. First, this tests only the reliability-heterogeneity half of the thesis—workers differ in accuracy and confusion patterns, but everyone answers in the same ten-way format, so there is nothing to decode. The convention-encoding half is not exercised by this dataset; it is tested on real data in Section 6.13. Second, the gold labels here are the CIFAR-10 test labels; on genuinely ambiguous images the "correct" answer is itself contested, which is one more reason not to over-read small gaps.

An audit-repair note: every dataset statistic quoted in this section (514,200 judgments; 2,571 annotators; exactly 200 judgments per annotator; mean worker accuracy 0.949, computed as 0.9487) was recomputed from the raw CIFAR-10H release for this revision, and the Dawid–Skene baseline was re-implemented as genuine iterated EM (an earlier released version computed the label posterior once and never iterated it). The repair changes only the five-judgment DawidSkene figure (0.9846 → 0.9850); the full-regime predictions are identical, and no conclusion in this section moves.

6.13 Independent validation: mechanism, real conventions, failure map

The experiments above are the paper's own. As a post-repair check we ran a separate validation programme (PVI-REAL; code and outputs in `code/pvi_real`, `results/pvi_real`) against hypotheses fixed before the runs: (H1) real sources' behaviour varies by context, including sign flips; (H2) the scalar failure grows with measured heterogeneity; (H3) the decoder recovers conventions from support with quantifiable sample efficiency; (H4) the advantage is mediated by source identity and context (ablations must destroy it); (H5) the effect survives on real data.

Mechanism under controlled panels. We re-ran the paper's generative machinery as a virtual panel study: five regimes (low heterogeneity; fixed inversion; context conventions; a boundary mix;

emergent drifting conventions), 40 panels of six sources per regime, 40 shared query items, support nested at $K \in \{0, 2, 5, 10, 20, 40\}$, with Majority, both scalars, ContextMatchFreq, a per-context transition-matrix method (PerContextTM), the grid Bayes of Section 5, and an oracle. Controls behave as the theory requires: at low heterogeneity the scalar is best (log loss 0.094 vs. the decoder's 0.117) and under fixed inversion it wins outright (0.154/0.938), while Majority collapses (0.614 accuracy). Where conventions vary by context the decoder's advantage is large and dose-responsive: at K=20, log loss 0.269 vs. the scalar's 0.488 and accuracy 0.886 vs. 0.781; in the emergent regime, 0.327/0.848 vs. 0.687/0.666. The panel-level advantage regresses on measured heterogeneity with slope 0.73 [0.65, 0.81], and the slope survives within the context regime alone (1.14 [0.58, 1.60]), so it is not an artefact of contrasting regimes. Shuffling source identities destroys 366% of the advantage (the decoder falls *below* the scalar), shuffling contexts 215%, and a placebo context bit exactly the 75% attributable to the real second context (H4). The decoder closes 80% (context) to 91% (emergent) of the scalar-to-oracle log-loss gap by K=40, reaches the scalar's K=40 log loss by K=5 in the context regime, and loses to the scalar in only 3 of 40 context panels—the lowest-heterogeneity ones. A power analysis on these panels sizes a human replication at ≈15 panels per arm. This arm validates estimator behaviour and internal consistency; by construction it cannot validate the shape of real conventions.

Real conventions in human judgments (CrowdTruth). The CIFAR-10H study above cannot test the convention half of the thesis: its contexts are unrecorded and its format uniform. The CrowdTruth medical relation corpus [9] records both: crowd workers judged candidate cause/treat relations in medical sentences, with expert labels, across two microtasks (relation extraction, RelEx; direction construction, RelDir) and two annotation channels. Taking contexts as (relation family × channel) and a labelled support/query split of items, the premise is directly measurable (H1): among workers with ≥15 judgments in each family, the median absolute cross-family accuracy gap is 0.155, and 37% are above chance in one family and below it in the other—real sources do flip by context. We evaluate the paper's decoding principle in its scalable form, a context-conditioned log-odds aggregator whose per-(source, context) weights are shrunk toward the source's scalar weight, with one temperature per method fitted by cross-fitting on the support only (no query data is touched in fitting); the full grid Bayes is not applicable here because these contexts are item families, not the involutive family of Section 3. On RelEx (5,564 judgments, 497 items, 67 workers, 20 splits), the calibrated context decoder beats the identically calibrated scalar on every metric: log loss 0.395 vs. 0.430 (paired advantage +0.035 [0.028, 0.041]), accuracy 0.830 vs. 0.807 (+2.3 points [1.7, 3.0]), calibration error 0.087 vs. 0.094; Majority scores 0.719/0.682 and Dawid–Skene 1.62/0.774. Uncalibrated, both weighted methods are severely overconfident (ECE ≈ 0.145) and the context method *loses* on log loss—calibration is part of the method, not a patch. Effects are context-specific and are reported per context as a matter of policy: the RelEx gain is carried by treat items (+4.0 [3.3, 4.6]) with cause items null (−0.2 [−1.3, +1.0]). On RelDir (4,481 judgments, 751 items, 77 workers) the same frozen pipeline replicates with a larger effect—log-loss advantage +0.101 [0.098, 0.105], accuracy +3.4 points [2.6, 4.2], gains in both families (cause +4.6 [3.5, 5.7], treat +2.0 [1.0, 2.9])—while Dawid–Skene collapses below chance (0.374). Two scope notes: both tasks share one dataset family and domain (medical text), so a different-domain replication is still owed; and ChaosNLI, an obvious candidate, is unusable for any per-source study—its release contains no annotator identifiers, which we verified item by item.

Adversarial failure map. Four regimes probe the assumptions directly (40 panels each). (i) *Out-of-set conventions*: sources use the pattern parity $= 1 - a_2$, outside the candidate set; per-cell methods win (ContextMatchFreq log loss 0.266, PerContextTM 0.258, decoder 0.590, scalar 0.744)—the candidate set is load-bearing, though misspecification degrades the decoder to "better than scalar" rather than to uselessness. (ii) *Reliability-only context*: reliability varies by context with no flips; Majority is near-oracle (0.348/0.844) and context machinery only adds variance—the thesis's regime is sign-changing conventions, and outside it there is nothing to win. (iii) *Drift*: stability decays from 0.95 (support) to 0.60 (query); the decoder remains best on accuracy but at K=40 its log loss (0.461) slips behind ContextMatchFreq (0.421)—stale support buys confidence in parameters that have moved. (iv) *Mimic-then-flip*: sources behave as identity on support and flip on queries; every support-based method collapses to chance accuracy with log loss 1.3–2.9, against a query-truth oracle at 0.089/0.967. The stationarity assumption—conventions persist from support to query—is thus not a caveat but a priced precondition of the entire approach, ours included.

7 Discussion

Nonnegative source weighting is insufficient for the convention class studied here. Theorem 1 proves it; the ablation in §6.4 shows it. Anything that multiplies messages by nonnegative scalars inherits the failure, no matter how the scalars are estimated.

Signed scalars are necessary but not sufficient once conventions depend on context. Theorem 2 makes this precise: a source's Bayes log-odds contribution varies with context at any stability $\rho > 0$ and its effective sign flips with context once $\rho > 1/2$; no context-independent scalar can represent either. The signed-to-context-conditioned gap in the ablation is the empirical fingerprint.

Explicit context conditioning recovers much of the lost performance; Bayesian inference over the latent convention adds a smaller, decomposable margin. ContextConditioned reaches 0.7350 against Bayes's 0.7590. A 30-seed plug-in ablation (§6.4) decomposes Bayes's 7.2-point margin over the EM decoder of the same model: ≈ 4.7 points come from EMDecoder's decision rule discarding the fitted stability $\hat{\rho}$, ≈ 1.9 points from the grid posterior's better convention recovery than EM's point fit (0.640 vs. 0.393 of sources at $K = 10$), and ≈ 0.6 points from posterior averaging itself. Theorem 3 shows the posterior concentrates on the observational-equivalence class of the truth as $K \to \infty$; Experiment C traces that trajectory empirically. The diagnostic MetaAttn+Dec, given privileged decoded features, reaches 0.8240 against Bayes's 0.7590 and the oracle's 0.8263 (paired $\Delta = -0.0023 \pm 0.0008$ to the oracle)—so the learned models' bottleneck is representational: decoding, not capacity, accounts for nearly all of their gap. Three further stress tests sharpen the claim. First, the stronger learned

baselines (§6.4): a wider attention decoder (DeepAttn) does no better than the smaller one, while a self-attention meta-learner (TransformerMeta) finishes half a point below Bayes (0.7544 vs. 0.7590; paired $\Delta = -0.0046$, 95% CI $[-0.0070, -0.0022]$, $d_z = -0.71$) — a small but statistically detectable deficit under the same paired standard we apply elsewhere, and with 104,834 parameters to MetaAttn's 25,667 it is not capacity-matched to it. Capacity alone does not explain the gap; with enough expressiveness a learned sequence model can nearly match the Bayesian decoder, but none of the learned methods we tried beats it. Second, a discriminative source× context meta-learner (CtxLogReg, §6.4, 10 seeds) with the same support information reaches only 0.637, below both ContextConditioned and Bayes: expressiveness without the right inductive bias does not help at $K = 10$. Third, Experiment H breaks the model's independence assumption with correlated source errors and finds Bayes hurt no more than the baselines or the oracle. With matched support information and modest capacity, building in the generative structure beats learning it from scratch.

7.1 Limitations

The benchmark is synthetic: a controlled binary task, involutive conventions, uniform priors, deterministic contexts. Real systems have continuous messages, non-involutive mappings, and structured contexts. None of our theorems depend on binarity in an essential way. But the experiments do not test beyond it. Correlated sources are tested in Experiment H (no disproportionate harm), but the correlation form is one convenient choice, not an exhaustive study.

The convention family is finite, and the Bayesian method assumes the candidate set is known. Open-world systems would need to grow the set adaptively. Experiment G probes the incomplete-family case and finds smooth rather than catastrophic degradation, but a principled open-world extension is future work.

Posterior concentration is moderate at small K and sharpens with support size, just as Theorem 3 predicts.

Calibration is a second, weaker axis. The Bayesian posterior grows overconfident as conventions stabilise (ECE from 0.017 at $\rho = 0.5$ to 0.084 at $\rho = 1.0$), and the learned calibrator's probabilities degrade as groups grow (ECE from 0.14 at $N = 2$ to 0.34 at $N = 12$). Neither method's probabilities should be taken at face value without recalibration.

7.2 Design implications

First, *do not aggregate raw messages across heterogeneous sources*: averaging messages from agents with different conventions can do worse than chance. Second, *calibrate before aggregating*: keep a support set per source and combine calibrated log-odds rather than raw messages. Reliability heterogeneity is about *how often* a source is right; convention heterogeneity is about *how* it maps readings to messages. A source can be perfectly reliable and completely unreadable. A concrete setting where this bites is multi-agent deliberation: systems that synthesise judgments from persona-driven agents, where each role induces a stable mapping from evidence to expressed message, face exactly the convention problem studied here, and with no ground truth available to adjudicate disagreements. Decoding each persona's mapping before aggregating is what separates a forecast from a muddle; we offer this as motivation.

8 Limitations

Messages and truth are binary; the multi-class extension is not worked out, though nothing in the theorems looks essentially binary. The convention family is finite and fixed, and the aggregator assumes the candidate set is known; Experiment G starts on misspecification but an adaptive, open-world candidate set is future work. Sources were assumed conditionally independent given the truth; Experiment H breaks that assumption with correlated errors and finds the ranking preserved, but only one correlation structure was tested. Conventions are static; drifting codes would need online inference, which we do not attempt.

The core evaluation is synthetic, plus one controlled LLM-communication benchmark in which the models' private beliefs are supplied rather than elicited, real-human-judgment studies on CIFAR-10H and on two CrowdTruth tasks, a controlled-panel validation, and an adversarial suite (Section 6.13): no LLM debate, robot teams, or sensor fusion, and none of these is a substitute for those. An LLM debate experiment with emergent conventions—incentive-based personas rather than instructed rules, agents using their own knowledge—is designed and coded (`code/llm_debate.py`) as immediate future work. The CIFAR-10H study, moreover, exercises only the reliability-heterogeneity half of the thesis—its annotators share one response format, so there is no convention to decode. The CrowdTruth studies do exercise the convention half—workers' affirmation behaviour flips across relation families—but through a scalable log-odds decoder rather than the grid Bayes of the synthetic benchmark, on one dataset family in one domain (medical text); a different-domain replication, and a human panel experiment with induced conventions, remain future work (protocols in the repository). The adversarial suite also prices the static-convention assumption directly: under drift the decoder's accuracy degrades gracefully but stale support inflates its confidence, and under an abrupt convention switch between support and query every support-based method fails. The statistics are exploratory—100 seeds for Experiment A, 30 for Experiments B, C, E, and I, ten for the rest—enough to tell stable gaps from noise, not enough for confirmatory claims.

A recent deployment datapoint is suggestive of the same failure mode outside simulation. Anthropic's Sonnet 5.5 system card reports FrontierCode scores of 52.1% at `xhigh` effort and 46.2% at `max` effort [2]; the card reports the scores without attributing the gap to any mechanism. Secondary coverage of the release attributes the drop to the multi-agent code review invoked at maximum effort, whose review subagents sometimes timed out or made out-of-scope edits. We cite the datapoint as an analogy — more reviewers, worse outcome — not as evidence for our mechanism: it is not a controlled comparison, and it does not isolate the noise-not-signal effect our $\rho = 0.5$ result identifies.

The learned baselines were each run in a single, untuned training configuration, and the LLM benchmark's banked numbers were produced by a pre-fix harness version (labelled as such in §6.11); both carry that scope. On the method: the grid is coarse (ρ in steps of 0.1, r in steps of 0.05), though a resolution sweep moves

every headline number by at most 0.004 (Appendix B.2); the priors are flat, and posterior concentration at small K is moderate. Calibration is reported (NLL, Brier, ECE) but not corrected: the Bayesian posterior is overconfident at high convention stability, and we leave recalibration to future work. The claim is about decoding versus weighting, not optimal grid design.

9 Conclusion

Scalar reliability weighting cannot implement the Bayes rule in two regimes: nonnegative weighting cannot invert a source (Theorem 1), and no context-independent scalar can reproduce a source's context-varying Bayes log-odds contribution — failing on decisions as well once stability exceeds 1/2, where the effective sign itself flips with context (Theorem 2). The way out is to decode: our finite-grid Bayesian latent-convention aggregator outperforms the scalar baselines, matches or exceeds the learned calibrators we tested, and reaches within 4.3 points of the oracle at $N = 12$—37.5 points over naive averaging. It survives candidate-set misspecification gracefully and is not disproportionately hurt by correlated source errors. The independent validation programme (Section 6.13) carries the mechanism to real human judgments: where workers' behaviour measurably flips across recorded contexts, a calibrated context decoder beats the best calibrated scalar on accuracy, log loss and calibration across two tasks, and the adversarial suite marks the boundary—no edge when context only modulates reliability, and collapse for every support-based method when conventions switch between support and query. But decoding is not always worth its estimation cost: when conventions are unstable ($\rho = 0.5$) a population-average mapping beats per-source decoding, with the advantage vanishing as support grows (Experiment I)—a sharp, prescriptive boundary on when to decode. Its probabilities are well scored but overconfident at high convention stability. Source identity is not a reliability parameter bolted onto a message; it is part of the evidence-to-decision mapping.

AI-Generated Content Disclosure

Large language models were used as research and writing assistants in the preparation of this work, under the author's direction. Their roles were: drafting and copy-editing manuscript text; developing and debugging experimental and analysis code; assisting the audit pass, including locating internal inconsistencies and checking references against publisher records; and drafting analysis reports from computed outputs. Language models are additionally the *subject* of the benchmark in §6.11, which is a methodological use and is described there.

No result in this paper was generated by a language model. Every reported number was produced by executing the released code on the stated data, and the audit and validation passes recomputed the headline quantities from the raw outputs and raw datasets. No AI system is an author of this paper. The human author directed the work, reviewed and edited all AI-assisted text and code, and takes full responsibility for the paper's content, claims, and any remaining errors.

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A Deferred proofs

A.1 Equal-reliability optimum

Lemma 1. *For p honest and q inverted sources, all with reliability $r > 1/2$ and full stability ($\rho = 1$), the Bayes accuracy is*

$$A_{\text{Bayes}} = P\big(\text{Bin}(p+q, r) > \tfrac{p+q}{2}\big) + \tfrac{1}{2}P\big(\text{Bin}(p+q, r) = \tfrac{p+q}{2}\big),$$

while the best unsigned (nonnegative-weight) rule attains only

$$A_{\text{unsigned}} = P\big(\text{Bin}(p, r) > \tfrac{p}{2}\big) + \tfrac{1}{2}P\big(\text{Bin}(p, r) = \tfrac{p}{2}\big),$$

i.e. it optimally discards the inverted sources ($w_j = 0$).

Numerically, for $p = 2$, $q = 1$, $r = 0.8$: $A_{\text{Bayes}} = 0.896$ versus $A_{\text{unsigned}} = 0.80$; for $p = 3$, $q = 2$, $r = 0.8$: 0.942 versus 0.896. So nonnegative weighting does not merely underperform. Its optimal strategy is to throw away informative sources entirely. Thus, under this regime, the optimal unsigned rule discards the inverted sources rather than exploiting their information.

Proof. Work in $\{\pm 1\}$ encoding. With $\rho = 1$, honest sources emit $X_s = Z_s$ and inverted sources emit $X_{p+j} = -Z_{p+j}$, where the readings Z_i are conditionally independent given Y with $P(Z_i = Y) = r$. An unsigned rule computes $\text{sign}\big(\sum_{s=1}^{p} w_s Z_s + \sum_{j=1}^{q} v_j Z'_{p+j}\big)$ with $w_s, v_j \geq 0$, where $Z'_{p+j} = -Z_{p+j}$ satisfies $P(Z'_{p+j} = Y) = 1 - r < \frac{1}{2}$: each inverted message is *negatively* informative about Y.

Step 1: negative information is worse than noise. Fix an inverted source j and write the score as $v_j Z'_j + T$, where T collects all other terms. Conditional on $(T, Y) = (t_0, y)$,

$$P(\text{correct} \mid t_0, y) = (1-r)\,\mathbb{1}[\text{sign}(v_j y + t_0) = y] + r\,\mathbb{1}[\text{sign}(-v_j y + t_0) = y],$$

using $P(Z'_j = y \mid Y = y) = 1 - r$ and conditional independence of Z'_j from (T, Y) given Y. Replace Z'_j by a fair coin C_j independent of everything: the conditional accuracy becomes $\frac{1}{2}\mathbb{1}[\text{sign}(v_j y + t_0) = y] + \frac{1}{2}\mathbb{1}[\text{sign}(-v_j y + t_0) = y]$. The difference (coin minus Z'_j) is

$$(r - \tfrac{1}{2})\Big(\mathbb{1}[t_0 y > -v_j] - \mathbb{1}[t_0 y > v_j]\Big) \geq 0,$$

since $r > \frac{1}{2}$ and $-v_j \leq v_j$ (ties broken symmetrically preserve monotonicity). Hence replacing each inverted source's message by pure noise *weakly improves* every unsigned rule. Iterating over $j = 1, \ldots, q$,

$$A(w, v) \leq P\left(\text{sign}\left(\sum_{s=1}^{p} w_s Z_s + \sum_{j=1}^{q} v_j C_j\right) = Y\right),$$

with C_j i.i.d. fair coins independent of (Z, Y).

Step 2: noise cannot beat the honest-only Bayes rule. Write $S = \sum_{s=1}^{p} w_s Z_s$ and $N = \sum_{j=1}^{q} v_j C_j$. Since N is independent of (S, Y) and symmetric,

$$P(\text{sign}(S + N) = Y) = \mathbb{E}_N\big[P(\text{sign}(S + n) = Y)\big].$$

For each fixed n, $\text{sign}(S + n)$ is a deterministic function of the honest readings alone, so its accuracy is bounded by the honest-only Bayes accuracy: the majority vote over decoded readings, $P(\text{Bin}(p, r) > p/2) + \frac{1}{2}P(\text{Bin}(p, r) = p/2)$. Thus $A(w, v) \leq A_{\text{unsigned}}$ for all $w, v \geq 0$.

Step 3: achievability. Setting $v_j = 0$ and equal $w_s > 0$ gives exactly the honest-only majority vote, attaining the bound. The Bayes formula follows identically with all $p + q$ decoded readings. □

A.2 Proof of Theorem 2

Proof. Write $f_c(v, a) = v \oplus g(a)$ with $g = g_c$; the hypothesis is $g(a^{(1)}) = 0 \neq 1 = g(a^{(2)})$. Fix source j and a context a. The induced channel $Y \mapsto M_j$ is a binary symmetric channel: if $g(a) = 0$, both branches transmit R_j, so $\varepsilon_a := P(M_j \neq Y \mid Y, a) = 1 - r_j$; if $g(a) = 1$, the direct branch (probability $1 - \rho_j$) transmits R_j, erring with probability $1 - r_j$, and the convention branch (probability ρ_j) transmits $R_j \oplus 1$, erring with probability r_j, so $\varepsilon_a = (1 - \rho_j)(1 - r_j) + \rho_j r_j$. Neither error probability depends on the value of Y (each branch's error event is $\{R_j \neq Y\}$ or $\{R_j = Y\}$, both of Y-free probability), so the channel is symmetric and the Bayes log-odds contribution of message m in context a is $\ell_j(m, a) = (2m - 1) \log \frac{1-\varepsilon_a}{\varepsilon_a}$.

(a) In context $a^{(1)}$, $\varepsilon_a = 1 - r_j$ and the weight is $\lambda_j = \log \frac{r_j}{1-r_j} > 0$. In context $a^{(2)}$, ε_a is affine in ρ_j with slope $r_j - (1 - r_j) = 2r_j - 1 > 0$, hence increases strictly from $1 - r_j$ at $\rho_j = 0$ through $1/2$ at $\rho_j = 1/2$ to r_j at $\rho_j = 1$; the weight $\log \frac{1-\varepsilon_a}{\varepsilon_a}$ therefore decreases strictly from $+\lambda_j$ through 0 to $-\lambda_j$. Whenever $\rho_j > 0$ it differs from the $a^{(1)}$-weight λ_j, while a context-independent scalar assigns the same contribution in both contexts. Hence no such scalar equals source j's Bayes log-odds contribution in both.

(b) Let $\rho_j > 1/2$, so the $a^{(2)}$-weight $\ell_j^{(2)} := \log \frac{1-\varepsilon_{a^{(2)}}}{\varepsilon_{a^{(2)}}}$ is negative. Fix the remaining sources' messages and contexts with total Bayes log-odds S, $|S| < \min(\lambda_j, |\ell_j^{(2)}|)$ (attainable: source j alone gives

$S = 0$). Take $X_j = +1$. The total Bayes log-odds are $S + \lambda_j > 0$ on $x^{(1)}$ and $S + \ell_j^{(2)} < 0$ on $x^{(2)}$: the Bayes decisions differ while the centered message vector X is identical. Any context-independent rule is a function of X alone, so it assigns both observations the same prediction, contradicting agreement with the Bayes rule. (For $X_j = -1$ the signs reverse symmetrically.) □

A.3 Proof of Theorem 3

Proof. Support triples (M_i, a_i, Y_i) are i.i.d. across i. For any grid point $\theta \notin [\theta^*]$, the mean log-likelihood ratio

$$\frac{1}{K}\sum_{i=1}^{K} \log \frac{P(M_i, a_i, Y_i \mid \theta^*)}{P(M_i, a_i, Y_i \mid \theta)} \xrightarrow{a.s.} D_{\mathrm{KL}}(P_{\theta^*} \,\|\, P_\theta) > 0$$

by the strong law of large numbers, with strict positivity from the definition of the equivalence class. The grid is finite, hence so is the set of equivalence classes, and the minimum KL divergence to any class other than $[\theta^*]$ is strictly positive; write it as $\delta > 0$. The posterior odds of each foreign class against $[\theta^*]$ therefore decay as $\exp(-K(\delta + o(1)))$, uniformly over the finitely many classes, so $P([\theta^*] \mid \mathcal{D}_s) \to 1$ a.s.

For the predictive, write the log-odds at query (M_q, a_q) as $\log(N_K/D_K)$ with $N_K = \sum_\theta w_\theta\, p_1(\theta)$, $D_K = \sum_\theta w_\theta\, p_0(\theta)$, $w_\theta = P(\theta \mid \mathcal{D}_s)$, and $p_y(\theta) = P(M_q \mid Y = y, a_q, \theta)$. The query likelihood is the same conditional that appears in the support-triple distribution, so $p_y(\theta) = p_y(\theta^*)$ for every $\theta \in [\theta^*]$, for P_A-almost every query context a_q. Hence $N_K = p_1(\theta^*)P([\theta^*] \mid \mathcal{D}_s) + R_K$ with $|R_K| \le \max_\theta p_1(\theta)\,(1 - P([\theta^*] \mid \mathcal{D}_s)) \to 0$ a.s., and likewise for D_K. On the grid, $p_y(\theta) \in [0.05, 0.95]$, so the logarithm is continuous at the limit point and the predictive log-odds converge a.s. to $\log(p_1(\theta^*)/p_0(\theta^*))$: the oracle's log-odds. □

B Deferred experimental details

Table 7: The eight involutive conventions in the benchmark.

ID	Convention	Involutive	Context-dependent
0	identity	yes	no
1	NOT	yes	no
2	XOR with a_1	yes	yes
3	XOR with a_2	yes	yes
4	XOR with $(a_1 \wedge a_2)$	yes	yes
5	XOR with $(a_1 \vee a_2)$	yes	yes
6	XOR with $(a_1 \oplus a_2)$	yes	yes
7	XOR with $\neg(a_1 \wedge a_2)$	yes	yes

B.1 Posterior diagnostic

To check that the Bayesian inference actually identifies conventions rather than stumbling into good predictions by accident, we measure top-1 accuracy on the true convention and posterior mass concentration across 200 random sources with $K = 20$ (mean ± std over 10 seeds):

- Top-1 accuracy on the true convention: 0.828 ± 0.040.
- Fraction of sources with posterior mass exceeding 0.50 on the true convention: 0.722 ± 0.032.

These are diagnostics, not bounds. A diffuse posterior can still predict well if its errors cancel, and a concentrated posterior can predict badly if r or ρ are misestimated. What the numbers say is modest and reassuring: the posterior usually finds the right convention, and the predictive accuracy in §6.2 and §6.5 is consistent with that.

A natural follow-up is whether decisions survive convention misidentification. Conditioning on recovery across the same 200 sources per seed: decision accuracy is 0.831 when the top-1 convention is correct, but still 0.686 when it is wrong, well above chance. The posterior-predictive averages over the convention uncertainty rather than committing to a single (possibly wrong) convention, so the aggregator often gets the decision right even when it cannot name the code. Decision accuracy, not convention-identification accuracy, is what matters, and the Bayesian rule maintains decision accuracy well above chance even when identification fails.

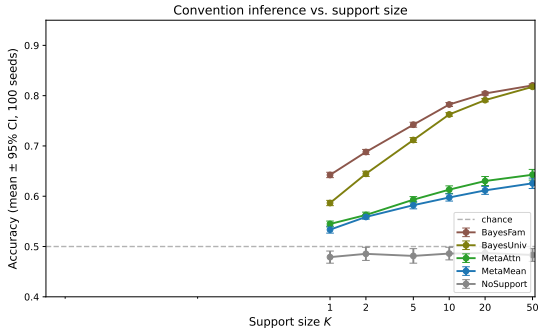


Figure 8: Accuracy on held-out conventions vs. support size K. Error bars show 95% CI over 30 seeds. Both Bayesian variants converge toward the per-query information ceiling (≈ 0.82); neural calibrators plateau near 0.64.

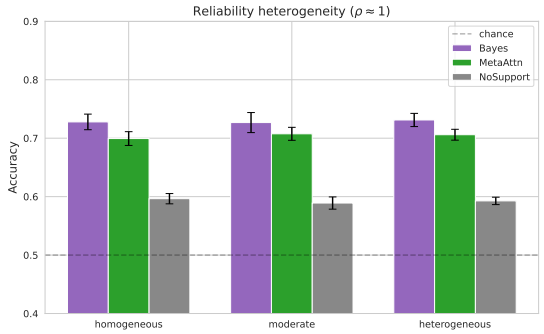


Figure 9: Robustness to reliability heterogeneity ($\rho \approx 1$, mean-matched $\mathbb{E}[r] = 0.80$). Error bars show ±1 std over 10 seeds. Accuracy tracks the mean reliability across spreads; heterogeneity per se does not degrade the Bayesian aggregator.

B.2 Robustness checks

Tables report mean ± std over ten seeds. 95% confidence intervals ($t_{0.975,9} = 2.262$), computed in torch-free NumPy drivers (`code/ci_check.py`, `code/sens_bayes.py`) that reproduce the paper's Bayes columns bit-identically on the default grid, confirm every headline gap: in the ρ sweep Bayes rises monotonically (0.625 ± 0.008 to 0.755 ± 0.009, adjacent intervals non-overlapping) and sits strictly above every baseline at $\rho = 1$; in the ablation Bayes (0.758 ± 0.011) is strictly below the oracle (0.825 ± 0.006) while the privileged MetaAttn+Dec (0.822 ± 0.006) is within noise of it; at $N = 12$ the +38.2 point consensus margin over naive averaging (0.955 ± 0.014 vs. 0.573 ± 0.044) and the 4.0 point oracle gap both hold; the misspecification drop (0.756 ± 0.009 to 0.685 ± 0.008) is real. The higher-seed reruns (Experiment A at 100 seeds; B, C, E at 30) reproduce every one of these headline values within the original intervals: $\rho = 1.0$ Bayes 0.7576 ± 0.0021, ablation Bayes 0.7590 ± 0.0044, $K = 50$ BayesFam 0.8204 ± 0.0029, $N = 12$ Bayes 0.9489 ± 0.0114 with a +37.5 point margin over naive averaging and a 4.3 point oracle gap. The one comparison that does *not* survive is ContextConditioned vs. MetaAttn (0.730 ± 0.010 vs. 0.722 ± 0.009): overlapping intervals, a point-estimate ordering only, as reported in §6.4. Grid resolution is not a driver: coarsening the grid 4.5× (96 cells) or refining it 3.5× (1496 cells) moves every headline number by at most 0.004. A likelihood-temperature sweep ($T \in \{0.5, 2.0\}$, flattening or sharpening the posterior toward or away from the uniform prior) preserves all orderings; only at the pathological $T = 5.0$ do magnitudes attenuate, and even there Bayes stays above the scalar baselines. The quantitative claims are calibrated to the stated uniform-prior model, and they are not prior-fragile.

Reproducibility

Everything is released: the finite-grid Bayesian aggregator, all neural baselines, the smoke tests, and a brute-force unit test (`code/test_predictive_fix.py`) that checks the posterior-predictive log-odds against explicit enumeration. From the repository root, `python code/smoke_test.py` verifies the core components in under ten seconds; `python code/upgrade_n100.py` reproduces the n100 runs in two invocations — `–only A –seeds 100` for Experiment A and `–only BCE –seeds 30` for Experiments B, C, and E — with calibration metrics and the stronger baselines, while `code/run_all_n10.py` reproduces the remaining ten-seed experiments (a few hours on a CPU-only machine). `code/analyze_n100.py` regenerates the confidence intervals and paired comparisons. The stability-boundary experiment (Section 6.3) is `code/boundary_experiment.py` (torch-free; 30 seeds over the $\rho \times K$ grid, ∼5 minutes on CPU), with figures from `code/make_figures_expI.py`. The real-human-judgment study is `code/real_world_cifar10h.py`; it reads the public CIFAR-10H raw file (`cifar10h-raw.csv` from `jcpeterson/cifar-10h`; point `CIFAR10H_CSV` at your copy, or use `data/cifar10h-raw.csv` in the repo) and writes `results/real_world_cifar10h.csv` plus `results/real_world_cifar10h_5judg.csv`, running in under a minute on a CPU. All randomness goes through a deterministic `SafeRNG` wrapper around Python's `random.Random`; PyTorch seeds are set before every model initialisation. The repository's `REPRODUCIBILITY.md` documents the environment, the per-experiment design, the baseline information budgets, and every correction applied during independent reproduction—including the posterior-predictive fix described in Section 5, which changed the estimator from a posterior expectation of log-odds to the exact predictive quantity. The code and data are provided as supplementary material; a persistent DOI will be added.

An audit-repair pass (October 2026) added `code/audit_fixes/`: the genuine Dawid–Skene EM now used by `code/real_world_cifar10h.py` (the earlier released version computed the label posterior once and never iterated it), the 30-seed plug-in ablation of §6.4 (grid-MAP and posterior-mean plug-ins, convention-recovery rates, the per-source log-odds scalar), the CtxLogReg support-size sweep, and exact parameter counts for the neural baselines; outputs are in `results/audit_fixes/`. All CIFAR-10H descriptive statistics were recomputed from the raw release in the same pass. The LLM benchmark's released harness now saves raw model outputs and logs parse-failure substitutions; its banked numbers predate those fixes (see §6.11). A validation programme in the same month added `code/pvi_real/`: the controlled-panel study and adversarial suite, the CrowdTruth RelEx/RelDir pipelines with the cross-fitted calibrated aggregators, and the deep-analysis scripts; every number in §6.13 recomputes from the summaries banked in `results/pvi_real/`. In the calibration outputs, NaiveMean and MajorityVote are both assigned the vote fraction as their probability; their hard decisions coincide by the sign argument of §6.7, so their probabilistic metrics are identical by construction.